Mitigating Extreme Class Imbalance in Credit Card Fraud Detection: A Hybrid AutoEncoder with Partial SMOTE and Weight-Swept Youden's J Threshold Optimization

Igdo Ragil Manuel
Faculty of Computer Science
Universitas Dian Nuswantoro
Semarang, Indonesia
111202314986@mhs.dinus.ac.id

Fauzi Adi Rafrastara
Faculty of Computer Science
Universitas Dian Nuswantoro
Semarang, Indonesia
fauziadi@dsn.dinus.ac.id

Wildanil Ghozi
Faculty of Computer Science
Universitas Dian Nuswantoro
Semarang, Indonesia
wildanil.ghozi@dsn.dinus.ac.id

Ramadhan Rakhmat Sani
Faculty of Computer Science
Universitas Dian Nuswantoro
Semarang, Indonesia
ramadhan_rs@dsn.dinus.ac.id

Gerald Guan Gan Goh
Centre of Excellence for Business Innovation and Communication
Multimedia University
Cyberjaya, Malaysia
gggoh@mmu.edu.my

L. Budi Handoko
Faculty of Computer Science
Universitas Dian Nuswantoro
Semarang, Indonesia
handoko@dsn.dinus.ac.id

***Abstract*—Credit card fraud detection is fundamentally challenged by extreme class imbalance (approximately 0.172% fraud ratio), where conventional SMOTE inflates False Positives and fixed thresholds are sub-optimal. This study proposes a hybrid framework integrating: (1) a 286-dimensional AutoEncoder feature space combining original attributes, latent representations, and MSE reconstruction error; (2) Partial SMOTE with a sampling ratio of 0.25 applied strictly post-extraction to suppress False Positives; and (3) a Weight-Swept Youden's J threshold optimization selected via cross-validation F1-Score. Five classifiers (XGBoost, Random Forest, CatBoost, LightGBM, MLP) are evaluated on the European Credit Card Fraud Dataset under an 80/20 stratified split with 5-fold CV. Results show AE+CatBoost achieves the highest Recall (94.90%) and AUC (97.90%), while AE+MLP attains the highest Accuracy (99.09%) and F1-Score (24.96%). All five models outperform the baseline study in AUC by 4.5 to 4.8 percentage points, validated by Bootstrap 95% CI.**

Keywords—AutoEncoder, Credit Card Fraud, Extreme Imbalance, Partial SMOTE, Weight-Swept Youden's J, Hybrid Features.

I. INTRODUCTION

The growth of digital payment systems has exposed financial institutions to increasing credit card fraud risk, driving the shift from conventional rule-based detection toward Machine Learning (ML) and Deep Learning (DL) approaches [1]. Designing robust fraud detection models is inherently constrained by extreme class imbalance in real-world transaction data. In benchmark datasets such as the European Credit Card Fraud dataset, fraudulent transactions represent merely 0.172% of total records [2]. This severe disparity causes conventional classifiers to favor the majority class, resulting in unacceptably high False Negative rates where actual fraudulent transactions go undetected [3].

To mitigate this, researchers frequently employ resampling techniques such as the Synthetic Minority Oversampling Technique (SMOTE) alongside Deep Learning feature extractors such as AutoEncoders (AE) to address complex imbalanced distributions [4]. While recent studies, such as the AE combined with Regularized Binary Cross Entropy (RBCE) [5], have reported high Detection Rates (Recall), a critical methodological gap persists in the current literature: the systematic omission of Precision and F1-Score metrics in highly imbalanced scenarios. Maximizing Recall without penalizing False Positives (FP) may present an incomplete picture of model performance, as the operational cost of elevated false alarm rates remains unquantified. A reverse-engineered estimation reconstructing the confusion matrix (TP, FP, FN, TN) from their reported Accuracy, Recall, and the 0.172% fraud ratio preserved through stratified K-fold evaluation reveals an implicit Precision of merely ~3.12% and F1-Score of ~6.04%. This implies that for every genuine fraud correctly flagged, dozens of legitimate transactions would be erroneously blocked, exposing an operational cost that prior evaluations relying solely on Accuracy and Recall fail to capture.

Furthermore, existing hybrid architectures often evaluate single-classifier configurations in isolation [5], [6] and apply SMOTE prior to cross-validation splitting, a flawed practice that introduces data leakage and artificially inflates evaluation metrics [7]. Additionally, the classification threshold is frequently set arbitrarily (e.g., default 0.5) or rigidly fixed, failing to dynamically balance the trade-off between fraud detection and false alarms.

To address these methodological gaps, this study proposes a comprehensive and operationally viable fraud detection framework integrating a hybrid AutoEncoder feature extraction mechanism, a leakage-free Partial SMOTE strategy, and a novel Weight-Swept Youden's J threshold optimization.

The main contributions are: (1) 286-dimensional hybrid AE feature space; (2) leakage-free Partial SMOTE (sampling ratio of 0.25); (3) Weight-Swept Youden's J with F1-based weight selection; and (4) Bootstrap 95% CI validation across five classifiers. Together, these contributions address the three core methodological gaps identified in existing literature within a single, reproducible, and operationally viable pipeline.

II. Related Works

A. AutoEncoders for Anomaly Detection

Prior studies have addressed credit card fraud detection through resampling techniques such as SMOTE [8], and unsupervised deep learning models, particularly AutoEncoders (AE), which learn compact representations of legitimate transaction patterns and can identify anomalous transactions through reconstruction error [9]. Recent analytical reviews indicate that both conventional machine learning and deep learning approaches remain widely used in credit card fraud detection, while hybrid and ensemble architectures have increasingly been explored to improve predictive performance [10]. These reviews also highlight that the effectiveness of fraud detection models remains strongly influenced by class imbalance, dataset characteristics, and the choice of evaluation metrics [10].

Sulaiman et al. [5] enhanced AE capabilities via a Regularized Binary Cross Entropy (RBCE) loss function with Bayesian hyperparameter optimization, achieving 95.13% Accuracy, 91.17% Detection Rate, and 93.15% AUC on the European credit card dataset; however, their evaluation relied on a fixed 0.95 threshold and omitted Precision and F1-Score, leaving the operational cost of False Positives unquantified. Sundaravadivel [3] independently validated AE-based fraud detection with alternative configurations, reporting competitive results on the same benchmark.

Recent Survey Perspective. Gaav et al. [10], in a systematic review of 40 credit card fraud detection studies, reported that supervised, deep learning, ensemble, and hybrid approaches dominate the field, that the European credit card dataset remains the most widely used benchmark, and that accuracy is reported far more consistently than Precision, Recall, or F1-Score. These findings reinforce the need for evaluation frameworks that jointly consider class imbalance, operational false alarms, and reproducible protocols rather than relying on accuracy alone.

B. Hybrid Classifier Architectures

Building on the hybrid AE-classifier paradigm, recent frameworks have combined unsupervised representation learning with supervised classifiers to improve fraud detection performance [11]. For instance, Prabha and Priscilla [12] combined an AutoEncoder with XGBoost [13] through probabilistic threshold shifting, achieving 90.4% recall on their evaluated benchmark. Ding et al. [6] proposed AEELG, pairing an AutoEncoder with LightGBM [14], highlighting the potential of combining representation learning with ensemble-based classifiers for high-dimensional fraud data. However, these approaches do not directly address the combined effect of extreme class imbalance, resampling strategy, and threshold selection within a unified leakage-controlled evaluation framework.

C. Identified Methodological Gaps

Despite these advances, several methodological gaps remain in existing credit card fraud detection studies.

First, threshold selection remains an important methodological challenge: Nata et al. [15] determine thresholds empirically, Prabha and Priscilla [12] explore threshold shifting, and Sulaiman et al. [5] employ a fixed 0.95 cut-off, but none provide a systematic mechanism for controlling the Recall-FPR trade-off, which is critical under extreme imbalance where small threshold changes substantially affect Precision, Recall, and FPR.

Second, operationally relevant metrics receive insufficient attention: Sulaiman et al. [5] report only Accuracy, Detection Rate, and AUC, while a reverse-engineered estimation of their model yields ~3.12% Precision and ~6.04% F1-Score, comparable to the ~3% Precision and 5% F1-Score reported by Nata et al. [15] indicating that a substantial proportion of generated fraud alerts may correspond to legitimate transactions.

Third, current literature lacks a unified, leakage-free framework: SMOTE is frequently applied prior to cross-validation, introducing data leakage, while aggressive resampling such as ADASYN [16] or full SMOTE (1:1) can make models overly sensitive, a pattern also observed in DDoS detection [17]. Moreover, prior studies, including the baseline study [6], report single point-estimate metrics without confidence intervals.

Fourth, the individual contributions of representation learning, class-imbalance handling, and threshold optimization are rarely isolated through systematic ablation, making it difficult to determine whether performance gains originate from the learned features, the resampling strategy, the decision threshold, or their interaction.

These gaps motivate the comprehensive, leakage-controlled framework and the controlled ablation comparisons presented in the following sections.

III. Research Methodology

The methodology integrates AutoEncoder-based feature extraction, a novel Partial SMOTE resampling strategy, and a dynamic Weight-Swept Youden's J threshold optimization. The overall research framework is illustrated in Fig. 1.

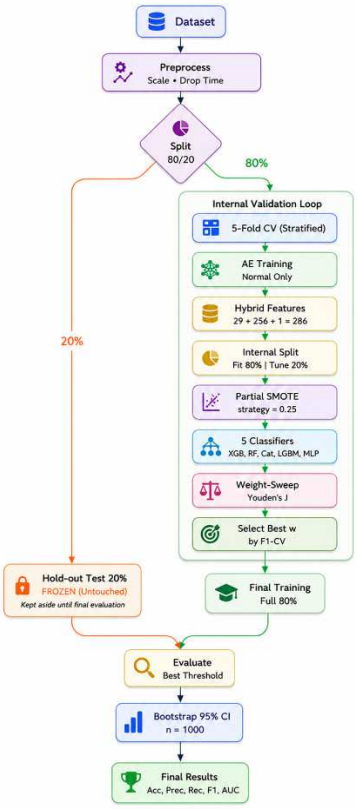


Fig. 1. Proposed research framework.

A. *Dataset Description*

The European Credit Card Fraud Detection Dataset [3] contains 284,807 transactions (492 fraud, 0.172%) from European cardholders collected over two days in September 2013, with 28 PCA-anonymized features (V1–V28) plus Time and Amount. The dataset characteristics are summarized in Table I. The severe class imbalance, with a fraud-to-legitimate ratio of 1:578, renders conventional accuracy-based evaluation misleading and necessitates specialized resampling and threshold optimization strategies, as addressed in subsequent sections.

TABLE I. CHARACTERISTICS OF THE CREDIT CARD FRAUD DATASET

Characteristic	Value
Dataset (Source, Period)	European Credit Card Fraud Dataset (ULB/Kaggle, Sept. 2013)
Total Transactions	284,807 - Legitimate: 284,315 (99.83%); Fraudulent: 492 (0.172%)
Features	28 PCA components (V1–V28), plus Time and Amount

The class distribution is illustrated in Fig. 2, confirming the severe imbalance between legitimate and fraudulent transactions.

The European Credit Card Fraud Detection Dataset was selected as the sole evaluation dataset because its extreme class imbalance provides a challenging and reproducible benchmark for evaluating fraud detection methods. The very small proportion of fraudulent transactions makes the dataset particularly suitable for investigating the effectiveness of class-imbalance handling, hybrid feature representation, and threshold optimization, which are the primary methodological components of this study. Furthermore, the use of this established benchmark facilitates comparison with existing credit card fraud detection studies that employ the same dataset.

Nevertheless, the use of a single benchmark dataset imposes limitations on the generalizability of the findings. The dataset represents transactions collected over a limited two-day period and provides anonymized PCA-transformed features, which may not capture all behavioral, temporal, and contextual characteristics available in real-world banking systems. Therefore, the reported results should be interpreted as evidence of the effectiveness of the proposed framework on this benchmark rather than as a guarantee of equivalent performance across different financial institutions, geographic regions, or evolving fraud patterns. Evaluation on additional and more diverse financial datasets remains an important direction for future work.



Fig. 2. Distribution of legitimate and fraudulent transactions in the European Credit Card Fraud Dataset.

This extreme imbalance can bias conventional classifiers toward the majority class, motivating the resampling and threshold optimization strategies discussed in Section III-C.

B. *Data Preprocessing*

The Time attribute was excluded, yielding 29 predictor variables (V1–V28 plus Amount), standardized via StandardScaler fitted exclusively on the training subset [18]. A global 80/20 stratified split was applied first, with the 20% hold-out set frozen until final evaluation. Within the 80% training set, stratified 5-fold cross-validation was employed; each fold was further partitioned into a Fit Set (80%) and Threshold-Tuning Set (20%) to prevent threshold optimization leakage. Partial SMOTE and classifier training were performed exclusively on the Fit Set, while the Threshold-Tuning Set retained its original 0.172% fraud ratio.

C. *AutoEncoder Architecture*

An AutoEncoder (AE) was trained exclusively on legitimate transactions (Class = 0) from the Fit Set within each CV fold, learning to reconstruct normal patterns such that anomalies produce larger reconstruction errors as an anomaly signal [19]. The proposed symmetric encoder–decoder architecture is summarized in Table II. The 29–512–256–512–29 architecture was designed as a symmetric encoder–decoder structure, with the input and output dimensions corresponding to the 29 standardized transaction features. The 256-dimensional latent layer provides a compressed representation of the input space while retaining sufficient capacity to model nonlinear relationships among transaction features. The symmetric 512-unit encoder and decoder layers provide additional representation capacity while maintaining a consistent reconstruction pathway. The architecture was selected as a practical configuration that balances representation capacity and model complexity, rather than as a claim of global architectural optimality.

TABLE II. ARCHITECTURE OF THE PROPOSED AUTOENCODER

Layer	Number of Nodes	Activation Function
Input Layer	29	-
Encoder Layer	512	ReLU
Latent Layer	256	ReLU
Decoder Layer	512	ReLU
Output Layer	29	Linear

Network parameters were optimized by minimizing the MSE reconstruction loss using the Adam optimizer[20]:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{x}_i)^2 \quad (1)$$

where x_i and $\hat{x}_i$ denote the original and reconstructed feature values, respectively, and n represents the number of input features. The complete hyperparameter configuration of the AutoEncoder is summarized in Table III.

TABLE III. AUTOENCODER HYPERPARAMETER

Hyperparameter	Value
Optimizer	Adam
Learning rate	0.0001
Epochs	50
Batch size	64
Early stopping patience	5
Validation split	0.1
L2 regularization	0.001

Early stopping and L2 regularization were applied to mitigate overfitting and improve generalization.

D. *Hybrid Feature Extraction*

The proposed hybrid feature representation combines three complementary sources: original standardized features, AE latent representations, and MSE reconstruction error, as summarized in Table IV. The hybrid feature vector is formulated as:

$$Xhybrid = [Xoriginal, Z, RE] \quad (2)$$

where $X_{original}$ denotes the original standardized features, $Z = f_{encoder}(X)$ is the latent representation extracted by the encoder, and RE is the reconstruction error computed using (2). This integration simultaneously preserves explicit transaction characteristics and encodes hidden anomaly signals learned from legitimate transactions. Partial SMOTE is subsequently applied exclusively to the Fit Set of this 286-dimensional hybrid space, while the Threshold-Tuning Set retains its original 0.172% fraud ratio.

TABLE IV. DIMENSIONS OF THE HYBRID FEATURE REPRESENTATIONS

Feature Component	Dimension
Original Features	29
Latent Features	256
Reconstruction Error	1
Total Hybrid Features	286

Table IV shows the original 29-dimensional feature space expanded to 286 dimensions, enriching downstream classifiers with both explicit transaction attributes and latent anomaly patterns from the AutoEncoder.

E. *Partial SMOTE Strategy*

The Synthetic Minority Over-sampling Technique (SMOTE) [21] is applied to prevent classifiers from biasing toward the majority class:

$$x_{new} = x_i + \lambda(x_{nn} - x_i) \quad (3)$$

where x_{new} denotes the generated synthetic sample, x_i represents a minority class instance, x_{nn} corresponds to one of its nearest minority neighbors, and $\lambda \in [0,1]$ is a random interpolation factor.

Unlike conventional full SMOTE (1:1 ratio), this study employs a Partial SMOTE strategy with sampling_strategy = 0.25, expanding the minority class to approximately 20% of the training data. The 0.25 ratio was selected as a controlled intermediate sampling level between the original highly imbalanced distribution and full class balancing. This configuration was intended to increase minority-class representation while avoiding the aggressive redistribution associated with full SMOTE. The effectiveness of this sampling ratio is subsequently examined through controlled comparisons with alternative imbalance-handling configurations, including no SMOTE and different SMOTE sampling ratios.

F. *Classification Models*

The effectiveness of the proposed hybrid feature representation was evaluated using five classification models: XGBoost (XGB) [13], Random Forest (RF)[22], CatBoost (CB)[23], LightGBM (LGBM)[14], and Multi-Layer Perceptron (MLP). These models were selected due to their demonstrated robustness in handling high-dimensional tabular data and their widespread adoption in financial fraud detection studies [6], [11], [12], [19], [24]. The selection of XGBoost is further supported by its reported effectiveness in other security-related classification tasks, including malware and ransomware detection [25]. Each classifier was trained using identical cross-validation folds and preprocessing procedures to ensure a fair comparison. The final hyperparameter configurations are summarized in Table V.

TABLE V. HYPERPARAMETERS OF THE CLASSIFICATION MODELS

Model	Hyperparameters
Random Forest	n_estimators = 200, max_depth = None, n_jobs = -1
XGBoost	n_estimators = 200, learning_rate = 0.05, max_depth = 6, min_child_weight = 5, subsample = 0.8, colsample_bytree = 0.8
LightGBM	n_estimators = 200, learning_rate = 0.05, max_depth = 6, subsample = 0.8, colsample_bytree = 0.8, is_unbalance = False
CatBoost	iterations=200, learning_rate=0.05, depth=6
MLP	hidden_layer_sizes = (256,128,64), activation = ReLU, solver = Adam, learning_rate_init = 0.001, max_iter = 200, early_stopping = True, validation_fraction = 0.1

Tree-based models used moderate complexity to balance performance and efficiency. Class weighting was omitted since imbalance was already addressed via Partial SMOTE, while MLP used early stopping (10% validation fraction) to mitigate overfitting.

G. *Threshold Optimization*

The default 0.5 threshold is sub-optimal for imbalanced fraud detection, failing to balance detection capability against false alarms. A Weight-Swept Youden's J [26] strategy was therefore employed to dynamically determine the optimal operating point for each model.

The threshold selection criterion was defined as follows:

$$J_{weighted}(t) = w \times TPR(t) - (1 - w) \times FPR(t) \quad (4)$$

where TPR and FPR are evaluated at threshold t, and w is a weighting parameter for fraud detection importance.

Unlike prior studies relying on a static weight, five discrete weight candidates ($w \in \{0.60, 0.65, 0.70, 0.75, 0.80\}$) were systematically swept, with the optimal weight and threshold selected via the highest F1-Score from 5-fold Stratified Cross-Validation, ensuring an objective, data-driven decision boundary. The search range was selected to place greater emphasis on fraud detection than on specificity, reflecting the asymmetric objective of identifying rare fraudulent transactions while still accounting for false alarms. The 0.05 increment provides a practical balance between search granularity and computational cost. The selected range is therefore treated as a controlled experimental search space rather than an assumption that a particular weight is universally optimal.

To prevent information leakage, this optimization is performed exclusively on the Threshold-Tuning Set (20% of the training fold, retaining its original 0.172% fraud ratio), with the resulting threshold applied to the corresponding testing fold and final hold-out set.

H. *Evaluation Metrics and Statistical Validation*

Performance was assessed using five metrics: Accuracy (5), Precision (6), Recall (7), F1-Score (8), and AUC (9). Given the extreme imbalance (0.172% fraud ratio), Accuracy alone can mask poor minority-class performance, as similarly shown in imbalanced malware detection [27]; Precision and F1-Score are therefore reported to quantify the operational cost of False Positives.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (5)$$

$$Precision = \frac{TP}{TP+FP} \quad (6)$$

$$Recall = \frac{TP}{TP+FN} \quad (7)$$

$$F1-Score = 2 \times \frac{Precision \times Recall}{Precision+Recall} \quad (8)$$

$$AUC = \int_0^1 TPR(FPR)d(FPR) \quad (9)$$

Greater emphasis is placed on Recall, F1-Score, and AUC over Accuracy as these better capture performance under extreme class imbalance; additionally, Bootstrap 95% CI [28] was performed using 1,000 resamples on the hold-out predictions, with intervals derived from the 2.5th–97.5th percentiles to confirm statistical robustness.

IV. Results and Discussion

A. *Experimental Setup*

Experiments used Python 3.11 (Scikit-learn, TensorFlow/Keras, XGBoost, LightGBM, CatBoost) on an NVIDIA GPU workstation, with a fixed random seed (42) for reproducibility.

B. *Classification Performance Comparison*

Table VI presents the comprehensive performance comparison of the five hybrid classifiers evaluated on the hold-out test set. The results reveal distinct operational trade-offs across the models, demonstrating that no single configuration universally dominates all metrics.

TABLE VI. Hold-Out Test Performance

Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	AUC (%)
AE+CatBoost	93.88	2.60	**94.90**	5.07	**97.90**
AE+LightGBM	97.31	5.63	92.86	10.62	97.69
AE+RandomForest	97.14	5.26	91.84	9.95	97.74
AE+XGBoost	98.49	9.46	90.82	17.13	97.74
AE+MLP	**99.09**	**14.55**	87.76	**24.96**	97.69

AE+CatBoost achieves the highest point-estimate Recall (94.90%) and AUC (97.90%); as shown in the Bootstrap CI analysis (Table VIII), the AUC improvement is statistically robust relative to the baseline, whereas the Recall advantage falls within overlapping confidence bounds. This comes at the cost of the lowest Precision (2.60%), resulting in substantial false alarms. In contrast, AE+MLP yields the highest Accuracy (99.09%), Precision (14.55%), and F1-Score (24.96%), proving its exceptional ability to suppress false alarms while maintaining detection sensitivity. Among tree-based models, AE+XGBoost offers a well-rounded profile with the best Accuracy (98.49%) and the second-highest F1-Score (17.13%), while AE+LightGBM and AE+Random Forest deliver the most balanced trade-offs (Accuracy >97%, Recall >91%), making them ideal for deployment scenarios that prioritize consistent performance across all metrics. Notably, all five models achieve AUC values exceeding 97.6%, confirming the discriminative superiority of the proposed 286-dimensional hybrid feature representation regardless of the downstream classifier used.

Despite the AUC and Recall improvements over the baseline, the absolute Precision achieved by the proposed models (2.60%–14.55%) implies a substantial residual false-alarm burden in practical deployment. AE+MLP, the best-performing configuration by Precision and F1-Score, correctly flags 86 of 98 fraud cases in the hold-out set at the cost of 505 false alerts, corresponding to approximately six false alarms per detected fraud. AE+CatBoost, despite achieving the highest Recall by detecting 93 of 98 fraud cases, generates 3,481 false alerts, corresponding to approximately 37 false alarms per detected fraud. These results indicate that the false-alarm burden should be considered carefully in operational settings, particularly where fraud analysts have limited investigation capacity.

Nevertheless, the observed false-positive rates should not be interpreted as an absolute indication of deployment infeasibility. In practical banking environments, fraud detection models can be used as risk-scoring or alert-prioritization mechanisms rather than as unconditional transaction-blocking systems. Under such an operating mode, the generated alerts can be ranked according to fraud risk and subjected to subsequent analyst review, allowing institutions to control the number of cases forwarded for investigation. The appropriate operating point would depend on institution-specific factors, including acceptable alert volume, analyst capacity, customer-impact tolerance, and the relative financial cost of missed fraud versus false alarms.

Accordingly, AE+MLP is preferable in scenarios where reducing the analyst workload and false-alarm volume is prioritized, whereas AE+CatBoost may be more appropriate when maximizing fraud coverage is the primary objective. The proposed framework should therefore be regarded as a configurable fraud-risk screening and decision-support mechanism rather than a universally deployable automatic blocking solution. Further validation using institution-specific transaction streams, operational alert volumes, fraud-loss costs, and analyst workload would be required before production deployment.

C. *Comparison with Previous Studies*

A comparison with AutoEncoder-based prior studies on the Kaggle credit card fraud dataset was conducted, prioritizing Detection Rate and AUC given the severe class imbalance, while Accuracy was included when reported. Both baseline studies were verified to use the same benchmark: Sulaiman et al. [5] and Nata et al. [15] were conducted on the identical European Credit Card Fraud Dataset (ULB/Kaggle) used in this study, ensuring a fair and directly comparable evaluation. y reconstructing the unreported metrics from these baselines, this comparison explicitly exposes the operational cost of false positives often overlooked in existing literature. As detailed in Table VII, the results demonstrate how the proposed framework significantly bridges this gap without sacrificing detection capabilities.

TABLE VII. COMPARISON WITH BASELINE METHODS

Method	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	AUC (%)
AE+RBCE+SMOTE [5]	95.13	~3.12	91.17	~6.04	93.15
Autoencoder anomaly-based [15]	94.98	3.00	85.00	5.00	93.13
AE+CatBoost (Proposed)	93.88	2.60	**94.90**	5.07	**97.90**
AE+LightGBM (Proposed)	97.31	5.63	92.86	10.62	97.69
AE+Random Forest (Proposed)	97.14	5.26	91.84	9.95	97.74
AE+XGBoost (Proposed)	98.49	9.46	90.82	17.13	97.74
AE+MLP (Proposed)	**99.09**	**14.55**	87.76	**24.96**	97.69

AE+CatBoost achieves the highest Detection Rate (94.90%) and AUC (97.90%), substantially outperforming Sulaiman et al. [5] (91.17% and 93.15%, respectively). More importantly, while the baseline study exhibits an estimated Precision of merely ~3.12% and F1-Score of ~6.04%, the proposed AE+MLP achieves 14.55% Precision and 24.96% F1-Score, representing a 4.7× and 4.1× improvement respectively. Furthermore, all five proposed models achieve AUC values exceeding 97.69%, representing an improvement of +4.5 to +4.8 percentage points over both baselines, confirming the superior discriminative capability of the proposed 286-dimensional hybrid feature representationThese results demonstrate the overall effectiveness of the proposed framework, while the individual contributions of Partial SMOTE and Weight-Swept Youden's J optimization are further examined through the ablation experiments described in Section IV-D.

Notably, the baseline study [5] used stratified K-fold cross-validation without a frozen hold-out set, whereas this study used a stricter, untouched 20% hold-out protocol.

TABLE VIII. BOOTSTRAP 95% CONFIDENCE INTERVAL

Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)	AUC (%)
AE+CatBoost	93.88 [93.68–94.09]	2.61 [2.10–3.16]	94.81 [90.11–98.89]	5.07 [4.10–6.11]	**97.83 [95.57–99.56]**
AE+LightGBM	**97.31 [97.18–97.45]**	**5.63 [4.47–6.71]**	92.83 [87.50–97.70]	**10.61 [8.52–12.54]**	**97.66 [95.20–99.54]**
AE+Random Forest	**97.14 [96.99–97.28]**	**5.25 [4.25–6.31]**	91.94 [86.17–97.03]	**9.92 [8.11–11.83]**	**97.77 [95.40–99.64]**
AE+XGBoost	**98.49 [98.39–98.59]**	**9.49 [7.71–11.53]**	90.92 [85.19–96.08]	**17.17 [14.20–20.44]**	**97.81 [95.64–99.57]**
AE+MLP	**99.09 [99.02–99.17]**	**14.60 [11.89–17.33]**	87.69 [80.55–94.17]	**25.00 [20.91–29.16]**	**97.69 [95.46–99.35]**

Table VIII presents the Bootstrap 95% CI derived from 1,000 resamples of the hold-out predictions. Minor deviations between the point-estimates in Table VI and the bootstrap means in Table VIII (e.g., Recall of 94.90% vs. 94.81% for AE+CatBoost) are expected, as the latter reflects the average across 1,000 resampled iterations rather than a single hold-out evaluation. For AUC, all five models show CI lower bounds well above the baseline study (93.15%), confirming a robust improvement across all configurations. For Accuracy, Precision, and F1-Score, four of the five models (LightGBM, RF, XGBoost, MLP) show CI lower bounds exceeding the corresponding baselines (95.13% Accuracy, ~3.12% Precision, ~6.04% F1), confirming statistically robust improvements; AE+CatBoost, however, does not robustly exceed the baseline in these three metrics. For Recall, the CI lower bounds of all five models fall below the baseline study (91.18%), indicating that while point estimates are higher, the improvement in Recall is not statistically robust at the 95% confidence level. The relatively wide confidence intervals observed for Recall stem from the limited number of positive cases (n=98) in the hold-out test set, an inherent constraint of extreme class imbalance rather than model instability. Overall, AE+LightGBM, AE+RF, AE+XGBoost, and AE+MLP demonstrate statistically robust superiority over the baseline in four of five metrics (Accuracy, Precision, F1-Score, AUC), while the Recall advantage, though consistent across all five models, warrants validation on larger minority-class samples. These findings suggest that practitioners should prioritize Precision- and F1-oriented classifiers for deployment, reserving Recall-focused configurations such as AE+CatBoost for contexts prioritizing exhaustive fraud coverage over false-alarm cost.

D. *Ablation Study*

To isolate the individual contributions of the proposed framework's components, four controlled ablation experiments were conducted using the same leakage-free protocol as the main pipeline, using 3-fold stratified cross-validation due to the higher computational cost of repeated AutoEncoder retraining. Because extreme fraud scarcity (approximately 70-130 cases per fold) makes threshold-dependent metrics highly sensitive to fold-level sampling variation, consistent with the wide Bootstrap confidence intervals for Recall in Table VIII, the threshold-independent AUC is emphasized as the primary indicator of stability across configurations.

1) SMOTE Sampling Ratio Sensitivity: Table IX compares five sampling configurations (No SMOTE, and SMOTE ratios of 0.10, 0.25, 0.50, and 1.00) under a fixed threshold weight (w = 0.75) to isolate the effect of the resampling ratio alone.

TABLE IX. SMOTE SAMPLING RATIO SENSITIVITY

Model	None	0.10	0.25	0.50	1.00
AE+CatBoost	2.85	28.37	2.55	3.57	2.88
AE+LightGBM	4.01	1.63	2.23	1.64	2.57
AE+MLP	23.24	27.66	19.93	10.51	30.46
AE+Random Forest	12.56	5.89	10.16	3.82	2.77
AE+XGBoost	4.13	3.19	2.29	2.79	2.37

No single sampling ratio is uniformly F1-optimal across all five classifiers, reflecting the sensitivity of threshold-based metrics to fold-level variation under extreme imbalance. AUC, however, remains comparatively stable (97.8%–98.3% for four of five classifiers), except for AE+LightGBM under No-SMOTE (94.75%), indicating that some minority-class oversampling benefits representation learning regardless of the specific ratio. The sampling_strategy = 0.25 configuration was therefore retained as a deliberate, classifier-agnostic compromise that avoids this AUC degradation while

avoiding the aggressive redistribution of full SMOTE (1.00), rather than as a per-classifier F1-optimal value.

2) Threshold Optimization Ablation: Table X compares the default 0.5 threshold against the proposed Weight-Swept Youden's J threshold, using identical Hybrid + Partial SMOTE (0.25) configurations for all five classifiers.

TABLE X. DEFAULT THRESHOLD VS. WEIGHT-SWEPT YOUDEN'S J

Model	Method	Prec. (%)	Rec. (%)	F1 (%)	AUC (%)
AE+CatBoost	Default	65.45	83.25	73.23	97.68
AE+CatBoost	Youden's J	34.16	86.57	35.42	97.68
AE+LightGBM	Default	74.66	81.22	77.79	97.76
AE+LightGBM	Youden's J	4.32	91.37	8.13	97.76
AE+MLP	Default	77.30	79.19	78.10	95.36
AE+MLP	Youden's J	14.62	87.07	20.50	95.36
AE+Random Forest	Default	87.49	79.70	83.39	97.13
AE+Random Forest	Youden's J	4.14	93.40	7.88	97.13
AE+XGBoost	Default	72.02	82.23	76.79	98.03
AE+XGBoost	Youden's J	3.06	92.89	5.90	98.03

This trade-off is consistent across all five classifiers: switching to Weight-Swept Youden's J increases Recall by 4 - 13 percentage points while reducing Precision by up to 80 percentage points, with AUC unaffected (only the threshold changes). This confirms threshold optimization is a controlled shift of the operating point toward higher fraud coverage rather than an improvement in the underlying model, directly quantifying the trade-off discussed in Section IV-B and supporting the framework's use as a configurable risk-scoring mechanism.

3) Hybrid Feature Representation Ablation: Table XI compares four feature representations the original 29 standardized features, the 256-dimensional AE latent space alone, their concatenation (285 dimensions), and the full proposed 286-dimensional hybrid representation under fixed Partial SMOTE (0.25) and default threshold, across all five classifiers.

TABLE XI. FEATURE REPRESENTATION ABLATION

Model	Original (29)	AE Latent (256)	Original+Latent (285)	Hybrid+RE (286, proposed)
AE+CatBoost	68.44	71.61	71.91	70.55
AE+LightGBM	73.01	75.87	76.88	78.22
AE+MLP	78.81	68.27	75.77	76.90
AE+Random Forest	83.11	82.96	83.36	83.09
AE+XGBoost	70.47	75.26	76.55	75.72

The benefit of the hybrid feature representation is classifier-dependent. Gradient boosting models show the clearest gains: AE+LightGBM improves by 5.2 points (73.01% to 78.22%) and AE+XGBoost by 5.3 points (70.47% to 75.72%). AE+CatBoost and AE+Random Forest show only small differences (within 3.5 and 0.4 points, respectively), while AE+MLP is the only classifier for which the original space outperforms the hybrid representation, plausibly reflecting greater susceptibility to the curse of dimensionality in a single end-to-end model without built-in feature selection. This indicates that the hybrid feature space benefits gradient boosting classifiers most, which constitute three of the five evaluated models.

4) Component Ablation: To isolate the combined contribution of the hybrid feature representation and Partial SMOTE prior to threshold optimization, Table XII compares two configurations across all five classifiers, both evaluated at the default 0.50 threshold: the original 29-dimensional feature space without resampling (Baseline), and the full proposed 286-dimensional hybrid representation with Partial SMOTE, sampling_strategy = 0.25 (Hybrid+SMOTE). The subsequent contribution of Weight-Swept Youden's J, applied on top of the Hybrid+SMOTE configuration, is examined separately in Table X.

TABLE XII. COMPONENT ABLATION

Model	Config.	Prec. (%)	Rec. (%)	F1 (%)	AUC (%)
AE+CatBoost	Baseline	94.50	77.92	85.38	97.80
AE+CatBoost	Hybrid+SMOTE	66.92	84.26	74.39	97.32
AE+LightGBM	Baseline	48.86	68.26	54.90	84.59
AE+LightGBM	Hybrid+SMOTE	77.81	82.49	80.06	97.02
AE+MLP	Baseline	86.14	79.95	82.90	96.31
AE+MLP	Hybrid+SMOTE	71.40	79.70	75.27	95.52
AE+Random Forest	Baseline	92.85	77.16	84.20	94.53
AE+Random Forest	Hybrid+SMOTE	87.06	79.95	83.32	97.04
AE+XGBoost	Baseline	93.24	76.15	83.77	97.84
AE+XGBoost	Hybrid+SMOTE	70.72	81.98	75.90	97.51

Baseline: Original features (29-dim) + No SMOTE + Threshold 0.50. Hybrid+SMOTE: Hybrid features (286-dim) + Partial SMOTE (0.25) + Threshold 0.50. Both configurations use the default threshold; the effect of Weight-Swept Youden's J applied on top of the Hybrid+SMOTE configuration is reported separately in Table X.

The effect of combining the hybrid feature representation with Partial SMOTE is classifier-dependent. AE+LightGBM shows the clearest gain (AUC: 84.59% to 97.02%, F1: 54.90% to 80.06%), consistent with its AUC degradation without resampling observed in Table IX. AE+Random Forest shows a moderate AUC gain (94.53% to 97.04%) with a smaller Precision reduction, while AE+CatBoost, AE+MLP, and AE+XGBoost show minimal AUC change (within 0.5 points) but a consistent Precision decrease, reflecting synthetic minority samples introduced at a threshold not yet adjusted for the resulting decision-boundary shift.

As shown in Table X, Weight-Swept Youden's J increases Recall by 4-13 percentage points at the cost of reduced Precision. Overall, threshold optimization primarily drives fraud detection sensitivity, while hybrid features and Partial SMOTE improve the underlying model performance.

V. Conclusion

This study proposed a leakage-free hybrid framework integrating AutoEncoder feature extraction, Partial SMOTE (sampling_strategy=0.25), and Weight-Swept Youden's J threshold optimization, evaluated across five classifiers with Bootstrap 95% CI validation on the hold-out test set (56,962 transactions; 98 fraud cases). AE+CatBoost achieved the highest Detection Rate (94.90%) and AUC (97.90%), while AE+MLP attained the highest Accuracy (99.09%) and F1-Score (24.96%), with all five models substantially outperforming the baseline study in AUC (+4.5 to +4.8 pp) and demonstrating statistically robust improvements under Bootstrap 95% CI analysis. AE+MLP achieved a 4.7× Precision and 4.1× F1-Score improvement over the baseline, demonstrating the effectiveness of Partial SMOTE in suppressing False Positives while maintaining competitive Recall.

The proposed framework demonstrates strong potential for real-world banking deployment. Future work includes exploring Variational AutoEncoder architectures [29] and SHAP-based interpretability, and validating the framework on larger, more diverse financial datasets to improve generalizability and recall estimate precision beyond the 98 fraud cases available in this hold-out set.

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